Chapter 17 After Tax Economic Analysis

- **Tax Terminology**
  - *Gross income* (*GI*) is the total income for the tax year from all revenue producing functions of the enterprise. E.g., sales revenues, fees, rent, royalties, sale of assets.
  - *Income tax* is the amount of money transferred from a company or an individual to government taxing agencies.
  - *Operating expanses* (*E*) includes all costs incurred in doing business. E.g., AOC, M&O, salaries, utilities, etc.
  - Operating expanses are *tax deductible*.
  - Depreciation, *D*, is also tax deductible.
  - *Taxable Income* (*TI*) is the part of *GI* which is taxed.
  - For companies, $TI = GI - E - D$.
  - *Tax rate* (*T*) is the percentage of *TI* owed in tax,

$$\text{Taxes} = TI \times T .$$

  - In the U.S. taxes are paid to both state government and federal government. The *effective tax rate*, T_e, is defined as

$$T_e = \text{state rate} + (1 - \text{state rate})(\text{federal rate}) .$$

    Then, Taxes = $TI \times T_e$.
  - *Net Profit After Tax* (*NPAT*) is the amount of *TI* which remains after paying tax,

$$NPAT = TI \times (1 - T) .$$

- **Tax rates in the U.S.**[1]
  - In the U.S., corporate tax is a *graduated tax*. That is, tax is paid based on brackets. Tax increases with bracket limits.

	Taxable Income		T - (%)
Braket	Braket Min	Bracket Max	Brkt. Rate
1	$0	$50,000	0.15
2	$50,000	$75,000	0.25
3	$75,000	$100,000	0.34
4	$100,000	$335,000	0.39
5	$335,000	$10,000,000	0.34
6	$10,000,000	$15,000,000	0.35
7	$15,000,000	$18,333,333	0.38
8	$18,333,333	Over 18,333,333	0.35

  - For example, a corporate with a TI = $80 K is taxed $50\times0.15+(75-50)\times0.25+(80-75)\times0.34$ = $15.45 K.
  - The *average tax rate* a corporate pays is (Taxes)/TI. For the corporate with TI = $80 K, this rate is 15.45 /80 = 19.31%.
  - U.S. individuals are also taxed in a graduated manner.

Tax Rate	TI Filing Single	TI Filing Married and Jointly
0.10	0-$7,000	0-14,000
0.15	$7,001-$28,400	$14,001- $56,800
0.25	$28,401- $68,800	$56,801- $114,650
0.28	$68,801- $143,500	114,651-174,700
0.33	$143,501 - $311,950	$174,701- $311,950
0.35	Over $311,950	Over $311,950

[1] Check www.irs.gov for more details.

- The TI for individuals is estimated differently from that of companies,

  $TI = GI$ – personal exemptions –itemized deductions .

- **Tax rates in Lebanon**
  - According to law 173, 2/14/2000, Lebanese individuals and companies are taxed in a graduated manner as follows.

Tax Rate	TI
0.04	0- 9 M LL
0.07	9 M LL - 24 M LL
0.12	24 M LL - 54 M LL
0.16	54 M LL - 104 M LL
0.21 for individuals 0.15 for companies	Over 104 M LL

  - For individuals, the TI is equal to the individual's GI minus a personal exemption of 7.5 M LL. Married individuals get an additional exemption of 2.5 M LL and 500 K LL per child.

- **Before-tax and after-tax cash flow**
  - *Cash flow before tax* ($CFBT$) is the net cash flow we have been considering,

$$CFBT_t = GI_t - E_t - P_t + S_t \,.$$

  - *Cash flow after tax* is $CFBT$ minus tax,

$$\begin{aligned} CFAT_t &= CFBT_t - TI \times T \\ &= GI_t - E_t - P_t + S_t - (GI_t - E_t - D_t)T \,. \end{aligned}$$

- **Effect of Depreciation Methods and Recovery Periods**
  - The effectiveness of a depreciation method in reducing taxes is measured by the present worth of taxes paid,

$$PW_{\text{tax}} = \sum_{t=1}^{n} [(GI_t - E_t - D_t)T](P/F, i, t),$$

    where n is the recovery period.
  - It can be shown that accelerated depreciation methods (DB and DDB) yield the least PW_{tax}.
  - It can also be shown that shorter recovery periods (n) lower PW_{tax}.
  - Note that the total undiscounted tax amount is the same for any recovery period provided and depreciation method that the same book value is depreciated in all methods.

- **Depreciation recapture and capital gain (loss) for companies**
  - The magnitude of the selling price (salvage value), S, of a company asset relative to its first cost, P, and book value, BV_t, at the sale time has tax implications.
  - If $S > P$ ($> BV_t$), then the company is taxed on the *capital gain*, $GC = S - P$.
  - If $S > BV_t$, then the company is taxed on the *depreciation recapture*, $DR = \min(S, P) - BV_t$.[2]

[2] The "min" term in the expression of DR handles situation where selling price exceeds first cost. In these cases, tax is applied to $GC = S - P$ and to $DR = P - BV_t$.

- If $S < BV_t$, then the company earns a tax saving equal to tax on the *capital loss*, $CL = BV_t - S$.[3]
- The taxable income in year t is now given

$$TI_t = GI_t - E_t - D_t + DR_t + CG_t - CL_t \,,$$

and cash flow after tax is

$$CFAT_t = GI_t - E_t - P_t + S_t - TI_t \times T \,.$$

- **After-tax economic analysis**
  - After-tax economic studies, with PW, AW, and ROR methods, are performed similar to before-tax studies. However, the analysis is based on $CFAT_t$.
  - After-tax replacement studies are performed similar to before-tax studies but with $CFAT_t$. Tax considerations due to depreciation, DR, and CG (CL) come into play.
  - An *approximate* after-tax analysis is done without estimating $CFAT_t$ by utilizing $CFBT_t$ with a *before-tax* MARR, where

$$\text{Before-tax MARR} = \text{After-tax MARR} / (1 - T) \,.$$

- **After-tax economic value added analysis**
  - The term *economic value added* (EVA) indicates the monetary worth added by an alternative to bottom line.
  - EVA is estimated as the net profit after tax minus the cost of invested capital. In year t,

$$EVA_t = NPAT - i \times BV_{t-1} = TI(1 - T) - i \times BV_{t-1} \,,$$

[3] This tax saving is used to offset taxes elsewhere in the company business.

where i is the cost of capital (i.e., company's interest rate or MARR) and BV_{t-1} is the book value estimated from depreciation methods.

FACT. *Comparing alternatives based on the AW of EVA_t is equivalent to comparing alternatives based on the AW of $CFAT_t$.*

- **International corporate tax rates**

TABLE 17–7 Summary of International Corporate Tax Rates*

Tax Rate Levied on Taxable Income, %	For These Countries
≥ 40	United States, Japan, Saudi Arabia
36 to < 40	Canada, Germany, South Africa
32 to < 36	China, France, India, Mexico, New Zealand, Spain, Turkey
28 to < 32	Australia, Indonesia, United Kingdom, Republic of Korea
24 to < 28	Russia, Taiwan
20 to < 24	Singapore
< 20	Hong Kong, Iceland, Ireland, Hungary, Poland

*Sources: Extracted from KPMG's Corporate Tax Rates Survey, January 2004, and from country websites on taxation for corporations.